

Journal of the Air & Waste Management Association

ISSN: 1096-2247 (Print) 2162-2906 (Online) Journal homepage: www.tandfonline.com/journals/uawm20

Comparison of Particle Light Scattering and Fine Particulate Matter Mass in Central California

Judith C. Chow, John G. Watson, Kihong Park, Norman F. Robinson, Douglas H. Lowenthal, Kihong Park & Karen A. Magliano

To cite this article: Judith C. Chow, John G. Watson, Kihong Park, Norman F. Robinson, Douglas H. Lowenthal, Kihong Park & Karen A. Magliano (2006) Comparison of Particle Light Scattering and Fine Particulate Matter Mass in Central California, Journal of the Air & Waste Management Association, 56:4, 398-410, DOI: 10.1080/10473289.2006.10464515

To link to this article: https://doi.org/10.1080/10473289.2006.10464515

Published online: 29 Feb 2012.

Submit your article to this journal

Article views: 717

View related articles

Citing articles: 3 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=uawm20

TECHNICAL PAPER

ISSN 1047-3289 *J. Air & Waste Manage. Assoc.* **56**:398–410
Copyright 2006 Air & Waste Management Association

Comparison of Particle Light Scattering and Fine Particulate Matter Mass in Central California

Judith C. Chow, John G. Watson, Kihong Park, Douglas H. Lowenthal, Norman F. Robinson
Desert Research Institute, Reno, NV

Kihong Park
Gwangju Institute of Science and Technology, Gwangju, Korea

Karen A. Magliano
California Air Resources Board, Sacramento, CA

ABSTRACT

Particle light scattering (B_{sp}) from nephelometers and fine particulate matter ($PM_{2.5}$) mass determined by filter samplers are compared for summer and winter at 35 locations in and around California's San Joaquin Valley from December 2, 1999 to February 3, 2001. The relationship is described using particle mass scattering efficiency (σ_{sp}) derived from linear regression of B_{sp} on $PM_{2.5}$ that can be applied to estimated $PM_{2.5}$ from nephelometer data within the 24-hr filter sampling periods and between the every-6th-day sampling frequency. An average of σ_{sp} = 4.9 m^2/g was found for all of the sites and seasons; however, σ_{sp} averaged by site type and season provided better $PM_{2.5}$ estimates. On average, the σ_{sp} was lower in summer than winter, consistent with lower relative humidities, lower fractions of hygroscopic ammonium nitrate, and higher contributions from fugitive dust. Winter average σ_{sp} were similar at non-source-dominated sites, ranging from 4.8 m^2/g to 5.9 m^2/g. The σ_{sp} was 2.3 m^2/g at the roadside, 3.7 m^2/g at a dairy farm, and 4.1 m^2/g in the Kern County oilfields. Comparison of B_{sp} from nephelometers with and without a $PM_{2.5}$ inlet at the Fresno Supersite showed that coarse particles contributed minor amounts to light scattering. This was confirmed by poorer correlations between B_{sp} and coarse particulate matter measured during a fall sampling period.

INTRODUCTION

Nephelometers[1-3] quantify the light scattered by particles by drawing them into an enclosed chamber, shining a bright light through the particle cloud, and measuring the amount of light scattered from its original direction. This particle light scattering (B_{sp}, in units of inverse megameters [Mm^{-1}]) is often correlated with mass concentrations of suspended particulate matter (PM, in units of $\mu g/m^3$). Using the relationship $B_{sp} = \sigma_{sp} \times$ PM, where σ_{sp} is the particle mass scattering efficiency (m^2/g), several researchers have derived empirical relationships between B_{sp} and PM that enable one to be estimated by measuring the other.[4-12] The value of σ_{sp} varies with particle size distribution, particle composition, and relative humidity (RH),[13] so it must be derived for different seasons, monitoring locations, and even for different samples. In addition, σ_{sp} is most consistent and physically meaningful when it is applied to the fine particulate matter ($PM_{2.5}$) size fraction under dry conditions. Most $PM_{2.5}$ particles have diameters comparable to the wavelength (λ) of the incident light, resulting in high σ_{sp}. Particles larger than 2.5 μm tend to scatter light less uniformly than smaller particles, and for most nephelometers a portion of the forward-scattered signal is truncated.[14-21] For these reasons, nephelometer B_{sp} is most accurately used as a surrogate for $PM_{2.5}$ mass concentrations, rather than for larger size fractions.

Nephelometers were deployed along with 24-hr $PM_{2.5}$ filter samplers during the California Regional PM_{10}/$PM_{2.5}$ Air Quality Study (CRPAQS)[22] to (1) determine σ_{sp} and its variability with location and season; (2) understand how $PM_{2.5}$ changes during the 24-hr filter sampling period; (3) evaluate $PM_{2.5}$ levels on days between the every-6th-day filter sampling frequency at most sites; and (4) estimate $PM_{2.5}$ levels at sites that did not have collocated $PM_{2.5}$ filter measurements. The combination of inexpensive nephelometers[23-25] and filter samplers[26] that can be located on rooftops and power poles provides the possibility of obtaining large amounts of spatial and temporal information at minimal cost. The methods illustrated here can be applied in a wide variety of locations.

In this study, relationships between B_{sp} and $PM_{2.5}$ mass are derived for 35 monitoring sites in central California. A predictability function is developed that can be used to estimate $PM_{2.5}$ mass from B_{sp}. Differences caused by site type, sampling location, sampling period, and ambient $PM_{2.5}$ levels are examined to determine when and where B_{sp} can be used as a surrogate for $PM_{2.5}$ mass.

IMPLICATIONS

Relatively inexpensive and portable nephelometers and $PM_{2.5}$ samplers can be used to deploy a large spatial air-monitoring network. Relationships between B_{sp} and $PM_{2.5}$ can be reliably established, but these differ by site type and sampling period. Nephelometers should not be used in the absence of some collocated $PM_{2.5}$ filter sampling at the same or similar measurement sites.

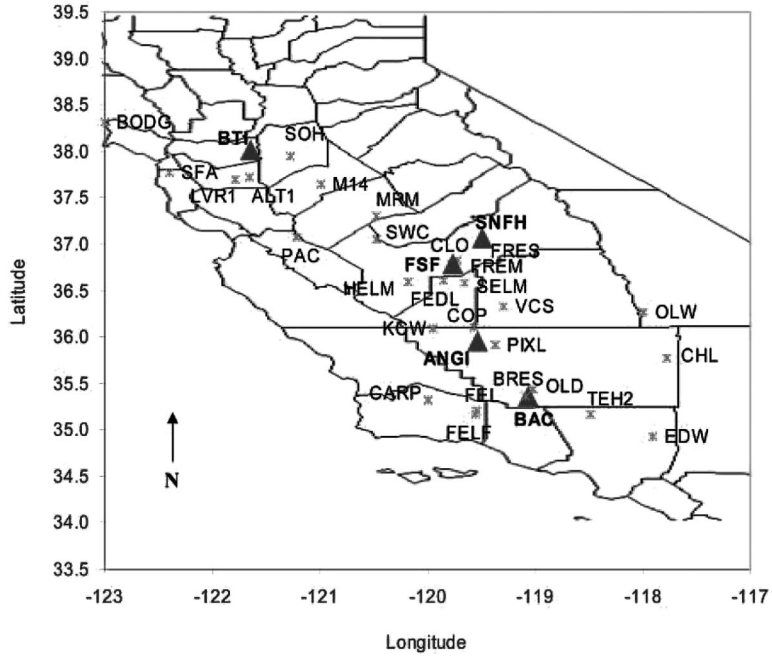


Figure 1. CRPAQS monitoring sites (5 anchor sites [▲] and 30 satellite sites [*]) with collocated Radiance Research M903 nephelometer and $PM_{2.5}$ filter samplers (Map not to scale).

Fine particle light scattering (B_{spf}) measured with $PM_{2.5}$ size-selective inlets is compared with total B_{sp} to evaluate the contribution from coarse particle scattering. Different nephelometers and filter sampling methods for B_{sp} and $PM_{2.5}$ mass are also compared.

METHODS

At the CRPAQS sites shown in Figure 1, Radiance Research (Seattle, WA) M903 nephelometers (λ = 530 nm) were collocated with Desert Research Institute (DRI; Reno, NV) $PM_{2.5}$ sequential filter samplers (SFS) at five anchor sites (Bethel Island [BTI], Sierra Nevada Foothills [SNFH], Fresno Supersite[27] [FSF], Angiola [ANGI], and Bakersfield [BAC]). These were also collocated with Airmetrics (Eugene, OR) battery-powered $PM_{2.5}$ MiniVol filter samplers at 30 satellite sites (Table 1). Most sites were located within California's San Joaquin Valley (SJV) except for the Bodega Bay (BODG), San Francisco (SFA), Olancha (OLW), China Lake (CHL), Tehachapi Pass (TEH2), and Edwards (EDW) sites. Table 1 summarizes locations (longitude, latitude), elevations, filter sampler types, site classifications and characterizations, and sampling periods.

Air was drawn into the nephelometer through an annular inlet (capped inlet tube with a 0.5-cm annulus) with a smart heater. The smart heater consists of a tube wrapped with heating tape that only applies heat when the RH at the outlet exceeds 72%. This heating reduces the enhancement of B_{sp} caused by water uptake of hygroscopic particles under high RH, while minimizing the evaporation of volatile material such as secondary ammonium nitrate (NH_4NO_3).[23,24]

The 24-hr (midnight-to-midnight) filter samples were collected every 6th day over a 14-month period (December 2, 1999, to February 3, 2001). The 24-hr SFS samples were taken at the FSF, ANGI, and BAC sites, starting on December 2, 1999, and at the BTI and SNFH sites, starting on December 2, 2000. Winter intensive operating periods (December 15, 2000 to February 3, 2001) included 5-times-per-day sampling for 15 days at the five anchor sites and 24-hr samples for 13 days at 25 satellite sites. The B_{sp} data were averaged over the 24-hr sampling periods for comparison.

At FSF, Radiance nephelometers with and without a $PM_{2.5}$ size-selective inlet were collocated to determine the extent to which coarse particles affect B_{sp}. An Optec NGN-2 ambient temperature low-truncation nephelometer (Lowell, MI, λ = 550 nm) measured total B_{sp}, including the portion caused by hygroscopicity and coarse particles. Non-integrating nephelometers included the TSI DustTrak (DT8520; Shoreview, MN, λ = 780 nm) and the GreenTek photometer (GT640A; Atlanta, GA, λ = 780 nm). These measure forward scattering at longer wavelengths and are more sensitive to coarse particles than integrating nephelometers that measure side-scatter at shorter wavelengths. Although the DustTrak and GreenTek units measure light scattering, they internally apply a σ_{sp} and give mass per unit volume outputs. Hourly average $PM_{2.5}$ and PM_{10} were measured at FSF with Met One (Grants Pass, OR) beta attenuation monitors (BAMs) without water vapor denuders and Rupprecht & Patashnick (Albany, NY) tapered element oscillating microbalances (TEOMs) operated at 50 °C. Also at FSF, $PM_{2.5}$ filter samples were taken with SFS, MiniVol, and Federal Reference Method (FRM; RAAS 100; Anderson Instruments, Smyrna, GA) samplers.

Nonweighted least-squares regression of B_{sp} on $PM_{2.5}$ mass was used to estimate σ_{sp} as the regression slope. A multiple linear regression was conducted when B_{sp}, $PM_{2.5}$, and $PM_{10-2.5}$ ($PM_{10} - PM_{2.5}$) mass were available.

Table 1. Central California sampling sites with collocated B_{sp} and $PM_{2.5}$ mass measurements.

Site Code	Site	Longitude	Latitude	Elevation (m)	$PM_{2.5}$ Filter Sampler	Site Characteristics	Sampling Period (data available)
ACP	Angels Camp	−120.491	38.006	373	MiniVol[a]	Intrabasin gradient	Nov. 30, 2000 to Jan. 31, 2001
ALT1	Altamont Pass	−121.660	37.718	350	MiniVol	Interbasin transport	Jan. 28, 2000 to Jan. 31, 2001
ANGI[c]	Angiola Trailer—ground level	−119.538	35.948	60	SFS[b]	Intrabasin gradient, vertical gradient, visibility	Feb. 1, 2000 to Feb. 3, 2001
BAC[c]	Bakersfield—5558 California Street	−119.063	35.357	119	SFS	Community exposure, visibility	Jan. 6, 2000 to Feb. 3, 2001
BODG	Bodega Bay	−123.073	38.319	17	MiniVol	Boundary/background	Dec. 23, 1999 to Jan. 31, 2001
BRES	Residential area near BAC with woodburning	−119.084	35.358	117	MiniVol	Source—woodburning	Nov. 29, 2000 to Jan. 31, 2001
BTI[c]	Bethel Island	−121.642	38.006	2	SFS	Interbasin transport	Dec. 1, 2000 to Feb. 3, 2001
CARP	Carrizo Plain	−119.996	35.314	598	MiniVol	Intrabasin gradient, visibility	Jul. 1, 2000 to Jan. 31, 2001
CHL	China Lake	−117.776	35.774	684	MiniVol	Visibility, desert environment outsite of SJV	Feb. 17, 2000 to Jan. 31, 2001
CLO	Clovis-908 N Villa Avenue	−119.716	36.819	108	MiniVol	Community exposure	Dec. 12, 2000 to Jan. 31, 2001
COP	Corcoran-Patterson Avenue	−119.566	36.102	63	MiniVol	Community exposure	Oct. 9, 2000 to Jan. 31, 2001
EDI	Edison	−118.957	35.350	118	MiniVol	Intrabasin gradient	Dec. 5, 2000 to Jan. 31, 2001
EDW	Edwards Air Force Base	−117.904	34.929	724	MiniVol	Intrabasin gradient, visibility, Desert environment outsite of SJV	Feb. 10, 2000 to Jan. 31, 2001
FEDL	Feedlot or Dairy	−119.855	36.611	76	MiniVol	Source—dairy	Jul. 7, 2000 to Jan. 31, 2001
FEL	Fellows	−119.546	35.203	359	MiniVol	Source—oilfields	Feb. 1, 2000 to Jan. 31, 2001
FELF	Foothills above Fellows	−119.557	35.171	512	MiniVol	Source—oilfields Intrabasin gradient	Mar. 18, 2000 to Jan. 31, 2001
FREM	Fresno-motor vehicle	−119.783	36.780	96	MiniVol	Source—motor vehicle	Jan. 10, 2000 to Jan. 29, 2001
FRES	Residential area near FSF with woodburning	−119.768	36.783	97	MiniVol	Source—woodburning	Jan. 30, 2000 to Jan. 31, 2001
FSF[c]	Fresno-3425 First Street	−119.773	36.782	97	SFS	Community exposure, visibility	Jan. 21, 2000 to Feb. 3, 2001
HELM	Helm-Central Fresno County	−120.177	36.591	55	MiniVol	Intrabasin gradient	Nov. 30, 2000 to Jan. 31, 2001
KCW	Kettleman City	−119.948	36.095	69	MiniVol	Intrabasin gradient	Nov. 27, 2000 to Jan. 31, 2001
LVR1	Livermore-793 Rincon at Pine	−121.784	37.688	138	MiniVol	Interbasin transport	Nov. 19, 2000 to Jan. 31, 2001
M14	Modesto 14th Street	−120.994	37.642	28	MiniVol	Community exposure	Nov. 11, 2000 to Jan. 31, 2001
MRM	Merced-Midtown	−120.481	37.308	53	MiniVol	Community exposure	Dec. 1, 2000 to Jan. 31, 2001
OLD	Oildale-Manor	−119.017	35.438	180	MiniVol	Community exposure	Nov. 28, 2000 to Jan. 31, 2001
OLW	Olancha-Walker Creek Rd	−117.993	36.268	1124	MiniVol	Background, desert environment outside of SJV	Feb. 17, 2000 to Jan. 31, 2001
PAC	Pacheco Pass	−121.222	37.073	452	MiniVol	Interbasin transport	Feb. 3, 2000 to Jan. 31, 2001
PIXL	Pixley Wildlife Refuge	−119.376	35.914	69	MiniVol	Interbasin gradient, rural	Jan. 26, 2000 to Jan. 31, 2001
SELM	Selma Airport	−119.660	36.583	94	MiniVol	Community exposure	Mar. 16, 2000 to Jan. 31, 2001
SFA	San Francisco-10 Arkansas	−122.399	37.766	6	MiniVol	Community exposure	Nov. 19, 2000 to Jan. 31, 2001
SNFH[c]	Sierra Nevada Foothills	−119.496	37.063	589	SFS	Intrabasin gradient, vertical gradient, visibility	Dec. 1, 2000 to Feb. 3, 2001
SOH	Stockton-Hazelton Street	−121.269	37.950	8	MiniVol	Community exposure	Nov. 29, 2000 to Jan. 31, 2001
SWC	SW Chowchilla	−120.472	37.048	43	MiniVol	Intrabasin gradient	Nov. 29, 2000 to Jan. 31, 2001
TEH2	Tehachapi Pass	−118.482	35.168	1229	MiniVol	Interbasin transport, visibility	Feb. 10, 2000 to Jan. 31, 2001
VCS	Visalia-North Church Street	−119.291	36.333	102	MiniVol	Community exposure	Nov. 30, 2000 to Jan. 31, 2001

[a]Battery-powered MiniVol samplers (Airmetrics, Eugene, OR) equipped with PM_{10} and $PM_{2.5}$ inlets (in series) or PM_{10} inlets operated at a 5 L/min flow rate; MiniVol samplers are shown to acquire $PM_{2.5}$ mass equivalent to $PM_{2.5}$ FRM compliance sampler; [b]Desert Research Institute (Reno, NV) SFS equipped with Bendix 240 cyclone $PM_{2.5}$ inlets and anodized aluminum nitric acid denuders operated at 113 L/min with 20 L/min flow rate through each sampling port; [c]Anchor sites; others are satellite sites.

RESULTS AND DISCUSSION

Spatial Variability of Light Scattering, $PM_{2.5}$ Mass, and Scattering Efficiencies

Comparisons of daily average B_{sp} and $PM_{2.5}$ mass grouped by site characteristics and season (winter and summer) are summarized in Table 2. As shown in Figure 2a, the 24-hr winter average B_{sp} varied by a factor of 21, from 14 Mm^{-1} at the OLW site to 299–303 Mm^{-1} at the Bakersfield residential (BRES), Edison (EDI), and Visalia (VCS) sites. The 24-hr winter average $PM_{2.5}$ mass varied by a factor of 46, from 1.3 $\mu g/m^3$ at the CHL site to 60.5 $\mu g/m^3$ at the Fresno roadside (FREM) site. Winter average $PM_{2.5}$ concentrations at urban sites (e.g., BRES, 53.9 $\mu g/m^3$; FRES, 57.4 $\mu g/m^3$; FREM, 60.5 $\mu g/m^3$) were twice the wintertime all-site average (28.9 $\mu g/m^3$). The lowest B_{sp} and $PM_{2.5}$ averages occurred at OLW (14 Mm^{-1}, 3.3 $\mu g/m^3$), CHL (15 Mm^{-1}, 1.3 $\mu g/m^3$), and EDW (26 Mm^{-1}, 5.4 $\mu g/m^3$); these were desert sites outside of the SJV.

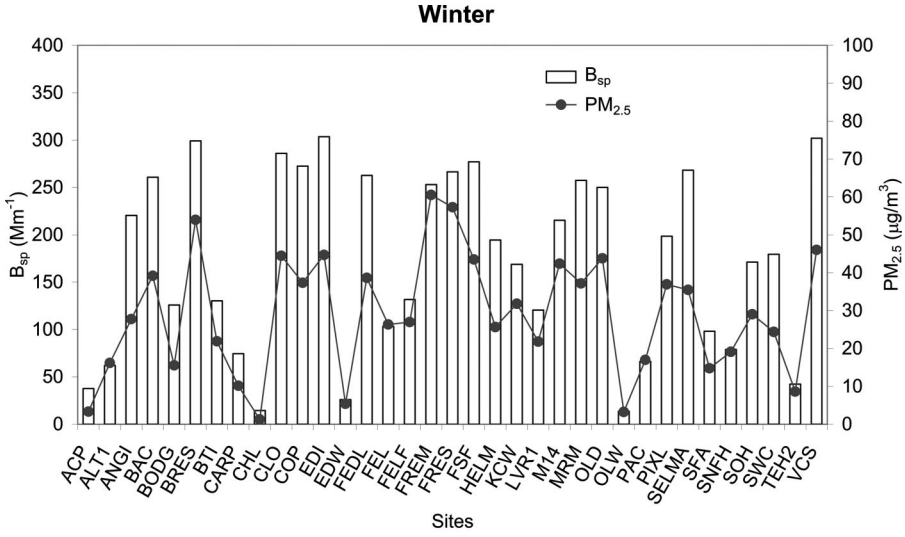
The spatial variability of σ_{sp} for the winter sampling period is shown in Figure 3. The desert sites, OLW, CHL,

Table 2. Comparisons of σ_{sp} derived from B_{sp} determined by a Radiance nephelometer and $PM_{2.5}$ mass determined by SFS (anchor sites) and MiniVol sampler (satellite sites).

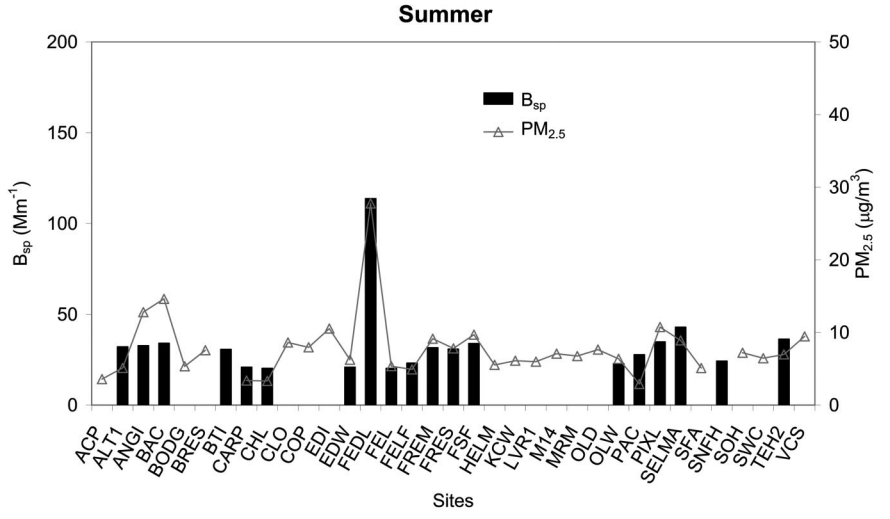
Site Code	Site Characteristics	Observables y	Observables x	Measurement Method y	Measurement Method x	Corr.[b] (r)	N[b]	Slope[b] ± Standard Error (m^2/g)	Intercept[b] ± Standard Error (Mm^{-1})	Winter Slope (m^2/g)	Winter Intercept (Mm^{-1})	Summer Slope (m^2/g)	Summer Intercept (Mm^{-1})
BAC[c]	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer[a]	SFS	0.91	366	6.1 ± 0.2	−28.7 ± 66.9	6.34	−1.11	1.40	13.89
CLO	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.85	10	5.2 ± 1.2	64.4 ± 76.7	5.16	64.43	NA[b]	NA
COP	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.89	18	4.5 ± 0.6	71.7 ± 26.3	4.03	107.31	NA	NA
FSF[c]	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	SFS	0.95	318	5.3 ± 0.1	−5.3 ± 3.3	5.01	13.45	2.98	4.82
M14	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.99	10	4.8 ± 0.2	6.8 ± 11.5	4.81	6.75	NA	NA
MRM	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.87	21	5.4 ± 0.7	78.1 ± 33.1	5.37	78.10	NA	NA
OLD	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.93	11	4.9 ± 0.6	18.1 ± 40.5	4.49	18.09	NA	NA
SELM	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.94	51	6.8 ± 0.3	−2.2 ± 9.2	5.83	46.36	4.11	4.73
SFA	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.79	11	6.0 ± 1.6	4.7 ± 36.2	5.99	4.65	NA	NA
SOH	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.91	11	5.8 ± 0.9	−7.6 ± 38.4	5.84	−7.61	NA	NA
VCS	Community exposure	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.93	11	5.2 ± 0.7	26.6 ± 44.9	5.17	26.60	NA	NA
Average at community exposure sites						0.91	838	5.7 ± 0.2	−9.8 ± 35.6	5.6	13.1	2.3	9.3
ALT1	Interbasin transport	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.95	60	4.9 ± 0.2	14.8 ± 2.8	4.78	18.14	NA	NA
BTI[c]	Interbasin transport	B_{sp}	$PM_{2.5}$	Nephelometer	SFS	0.96	60	6.2 ± 0.2	−2.9 ± 6.5	6.17	−2.29	NA	NA
LVR1	Interbasin transport	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.96	13	4.8 ± 0.4	25.3 ± 14.7	4.81	25.33	NA	NA
PAC	Interbasin transport	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.91	56	5.2 ± 0.3	16.7 ± 3.8	5.04	20.79	NA	NA
TEH2	Interbasin transport	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.68	41	5.0 ± 0.9	11.3 ± 8.8	4.90	4.71	3.69	5.79
Average at interbasin transport sites						0.89	230	5.3 ± 0.4	10.6 ± 5.7	5.2	11.5	3.69	5.79
ACP	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	NA	8	NA	NA	NA	NA	NA	NA
ANGI[c]	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	SFS	0.83	269	5.7 ± 0.2	−10.9 ± 6.4	6.82	8.84	0.24	31.03
CARP	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.89	20	6.3 ± 0.8	9.7 ± 8.1	5.48	9.67	NA	NA
EDI	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.94	10	4.4 ± 0.6	65.1 ± 43.8	4.42	65.10	NA	NA
HELM	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.87	11	5.0 ± 1.0	52.0 ± 27.8	5.03	52.05	NA	NA
KCW	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.95	10	4.4 ± 0.5	34.9 ± 20.1	4.38	34.95	NA	NA
PIXL	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.90	57	5.5 ± 0.4	8.3 ± 10.5	4.90	32.30	NA	NA
SNFH[c]	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	SFS	0.79	64	4.8 ± 0.5	15.5 ± 10.9	4.87	12.28	NA	NA
SWC	Intrabasin gradient	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.96	10	5.1 ± 0.5	40.9 ± 20.5	5.10	40.90	NA	NA
Average at intrabasin gradient sites						0.9	459	5.4 ± 0.3	1.6 ± 9.4	5.9	15.6	0.24	31.03
BODG	Boundary/background	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.69	22	4.9 ± 1.1	39.2 ± 19.2	4.75	51.00	NA	NA
BRES	Source-woodburning	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.97	11	4.5 ± 0.4	12.8 ± 30.4	4.52	12.82	NA	NA
FRES	Source-woodburning	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.98	55	5.5 ± 0.2	7.0 ± 6.6	5.23	10.28	3.18	6.11
Average at woodburning source sites						0.97	66	5.3 ± 0.2	7.9 ± 10.6	5.1	10.7	3.18	6.11
FEL	Source-oilfields	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.93	40	4.1 ± 0.3	6.2 ± 5.0	3.78	28.09	1.90	7.77
FELF	Source-oilfields	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.81	38	3.7 ± 0.4	3.8 ± 9.0	4.42	7.29	2.55	8.16
Average at oilfield sites						0.87	78	3.9 ± 0.3	5.1 ± 6.9	4.1	18.0	2.2	8.0
FREM	Source-motor vehicles	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.68	61	3.2 ± 0.5	39.5 ± 19.5	2.27	110.75	1.31	20.32
FEDL	Source-dairy	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.38	29	2.8 ± 0.3	106.4 ± 43.5	3.65	92.15	2.39	41.81
CHL	Desert sites outside of SJV	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.87	36	4.0 ± 0.4	5.4 ± 1.2	0.10	9.30	4.90	1.90
EDW	Desert sites outside of SJV	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.55	39	2.8 ± 0.7	10.0 ± 4.8	1.76	12.89	0.59	17.64
OLW	Desert sites outside of SJV	B_{sp}	$PM_{2.5}$	Nephelometer	MiniVol	0.97	40	4.1 ± 0.2	0.8 ± 1.7	0.10	9.61	4.50	0.54
Average at desert sites outside of SJV						0.80	115	3.6 ± 0.4	5.4 ± 2.6	0.7	10.6	3.3	6.8

[a]Radiance Research M903 nephelometer (Seattle, WA); [b]Results from linear regression between B_{sp} (y-axis) vs $PM_{2.5}$ mass (x-axis) for sampling periods given by Table 1; [c]Anchor sites.

and EDW, yielded low σ_{sp} of 0.1, 0.1, and 1.8 m^2/g, respectively. The OLW and CHL values are not accurate owing to the higher measurement uncertainties at the low concentration levels. According to the site characteristics shown in Table 2, average wintertime σ_{sp} are 4.8 m^2/g for background sites, 5.2 m^2/g for interbasin transport sites, 5.6 m^2/g for community exposure sites, and 5.9 m^2/g for intrabasin gradient sites. With the exception of σ_{sp} at residential sites (5.1 m^2/g) where home heating is believed to be a nearby source, lower σ_{sp} were determined at sites near sources, including motor vehicle (2.3 m^2/g), dairy farm (3.7 m^2/g), and oilfield (4.1 m^2/g). Most of the values are higher than the Interagency Monitoring of Protected Visual Environments (IMPROVE) dry chemical scattering efficiencies, indicating that the samples retain some liquid water, probably associated with NH_4NO_3, ammonium sulfate $[(NH_4)_2SO_4]$, and possibly with some of the organic material. The RH set point



(a)



(b)

Figure 2. Average B_{sp} and $PM_{2.5}$ mass concentration at CRPAQS sites in (a) winter (February 2000, December 2000, and January 2001) and (b) summer (June 2000 to August 2000). See Table 1 for site codes.

on the nephelometer smart heater should probably have been set at a level lower than 72% to remove more of the particle liquid water. Different amounts of liquid water probably cause some of the scatter observed in the B_{sp}/$PM_{2.5}$ comparisons.

Seasonal Differences for Light Scattering, $PM_{2.5}$ Mass, and Scattering Efficiencies

Particle compositions and size distributions in the SJV have a distinct seasonal pattern.[13,27-33] Chow et al.[28] showed that $PM_{10-2.5}$ dominates high PM_{10} levels during summer in the SJV, whereas $PM_{2.5}$ constitutes most of the high PM_{10} concentrations during winter. During winter, shallow surface layers form under cold conditions and enhance the accumulation of carbon particles from fresh emissions and secondary NH_4NO_3.[33] Coarse particles in the SJV are suppressed by periodic precipitation and fog. During CRPAQS, elevated $PM_{2.5}$ concentrations in winter accounted for more than 50% of the annual $PM_{2.5}$ level inside the SJV. That contribution was more pronounced in urban areas where fresh carbon emissions from vehicle exhaust and home heating can accumulate.[34]

Figures 2a and 2b also show that average B_{sp} and $PM_{2.5}$ mass were higher during winter than summer. In summer, the dairy farm (FEDL) site had the highest $PM_{2.5}$ (27.9 $\mu g/m^3$). Elevated $PM_{2.5}$ levels at the FEDL site were also found during winter (38.6 $\mu g/m^3$) and occurred throughout the year. These high levels were not reflected at nearby monitors, indicating that the zone of influence for this source was small.[34]

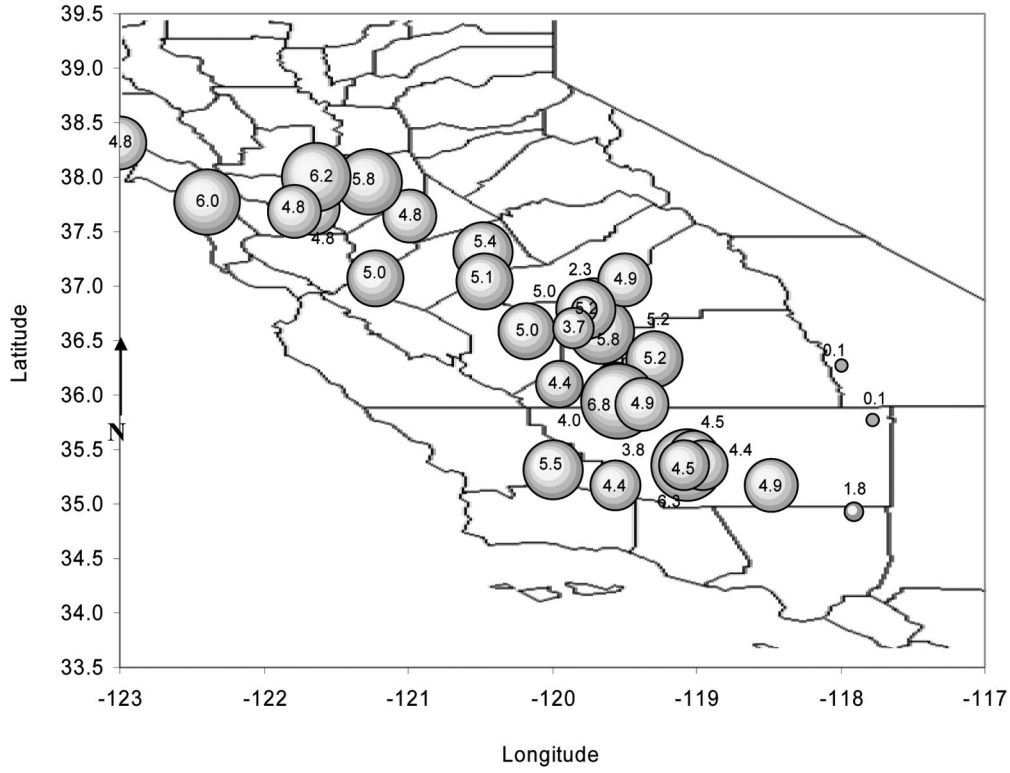


Figure 3. Spatial variation of σ_{sp} (m^2/g) averaged over winter sampling periods (the diameter of the circle is proportional to σ_{sp} except for the easternmost desert sites).

Winter and summer σ_{sp} as a function of $PM_{2.5}$ mass is shown in Figure 4. The average σ_{sp} was 2.6 m^2/g in summer and 4.6 m^2/g in winter, consistent with differences in particle composition, particle size, and RH that affect σ_{sp}. Suspended dust events are more common during summer, owing to drier conditions and more agricultural activity. Higher summertime σ_{sp} at the OLW and CHL sites is influenced by the outflow of $PM_{2.5}$ from the SJV and from the South Coast Air Basin that provides a larger fraction of $PM_{2.5}$ than during winter, when such outflow is suppressed.[35,36]

Daily average B_{sp} and $PM_{2.5}$ at the regional-scale ANGI and urban-scale BAC sites in winter and summer are shown in Figures 5a and 5b, respectively. Both sites display a similar seasonal dependence of B_{sp} on $PM_{2.5}$ (i.e., higher scattering efficiency and correlation during winter

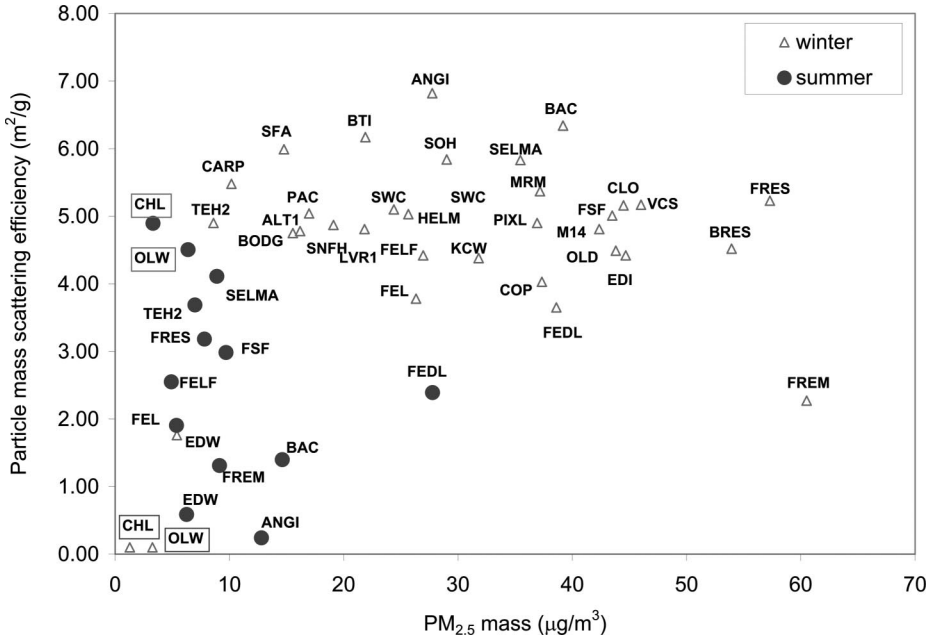


Figure 4. σ_{sp} during winter and summer.

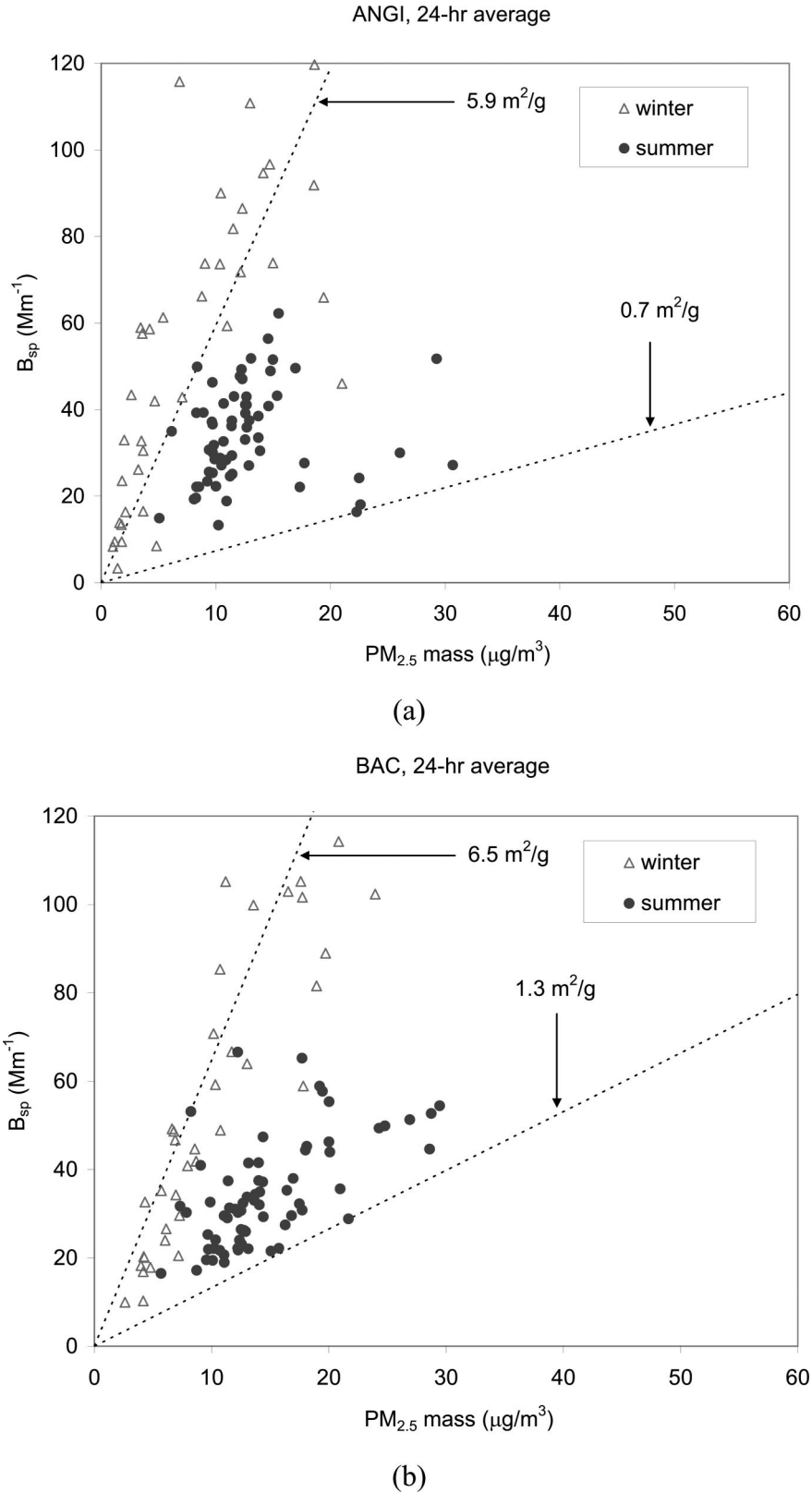
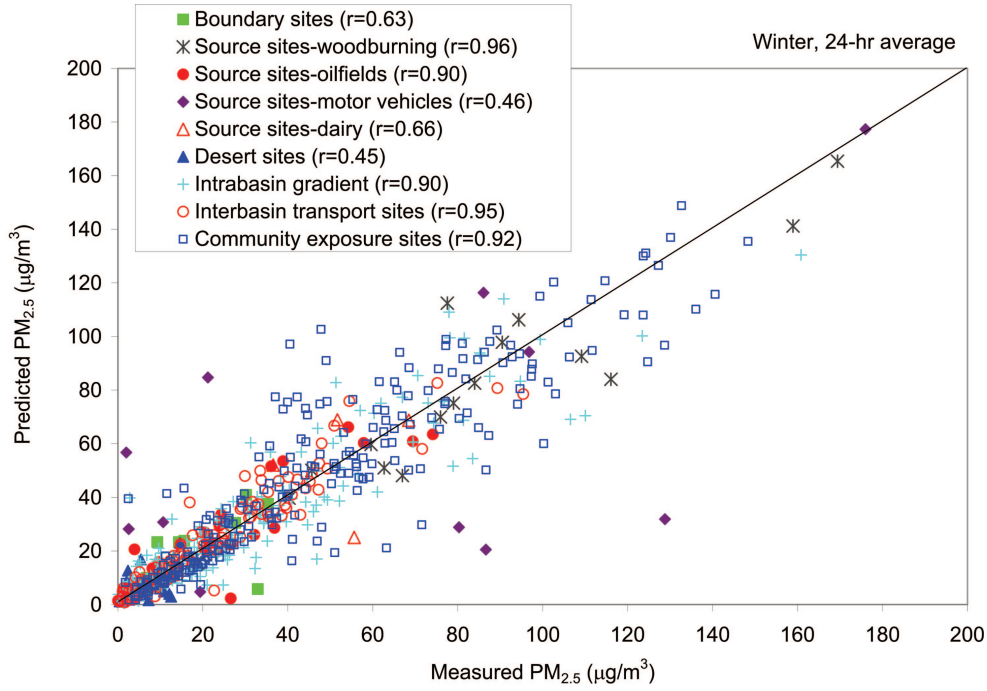


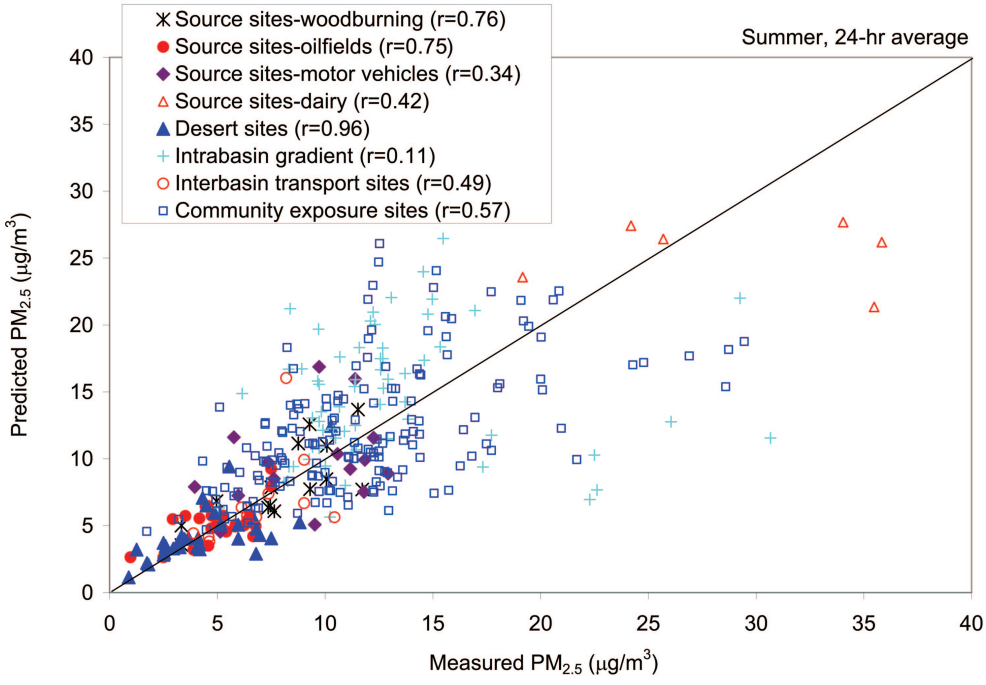
Figure 5. Daily average B_{sp} and $PM_{2.5}$ mass at the (a) ANGI and (b) BAC sites during winter and summer.

than during summer). Summertime σ_{sp} varied widely from 0.7 m^2/g to 5.9 m^2/g at ANGI and from 1.3 m^2/g to 6.5 m^2/g at BAC. Wintertime relationships are much more consistent.

When an all-site and both-season average of σ_{sp} = 4.9 m^2/g is used to estimate $PM_{2.5}$ mass from B_{sp}, the slope between the estimated and measured $PM_{2.5}$ mass deviates from a 1:1 line, especially during the summer



(a)



(b)

Figure 6. Comparison of $PM_{2.5}$ mass estimated from B_{sp} and measured $PM_{2.5}$ mass at different types of sites during (a) winter and (b) summer using site-type specific σ_{sp}. Correlation coefficients (r) between estimated and measured $PM_{2.5}$ are given in the legend for each site type.

(slope = 0.36). This average B_{sp} does not accurately estimate $PM_{2.5}$ mass everywhere all the time. Figures 6a and 6b compare $PM_{2.5}$ estimated from B_{sp} with the measured $PM_{2.5}$ mass at different types of sites in winter and summer, respectively, showing closer agreement when site-type and seasonal σ_{sp} values are used. Community exposure and interbasin gradient/transport sites, as well as source-dominated home heating and oilfield sites, show correlations exceeding 0.9 during winter. Correlations are low during summer, with the

Table 3. Comparison of particle light scattering and $PM_{2.5}$ at the FSF for data acquired between September 1, 2000, and August 31, 2001.

Sampling period	Observables y	Observables x	Measurement methods y	Measurement methods x	y Average ± std (Mm^{-1})	x Average ± std (Mm^{-1})	Corr. (r)	N	Slope ± Standard Error (m^2/g)	Intercept ± Standard Error (m^2/g)
Winter[a]	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP[a]	BAM	334.7 ± 233.4	70.0 ± 47.6	0.96	1524	4.50 ± 0.03	10.51 ± 2.69
Summer[b]	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	BAM	21.9 ± 20.7	12.4 ± 8.2	0.84	1895	1.72 ± 0.03	0.41 ± 0.38
Winter	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	TEOM	334.7 ± 233.4	27.7 ± 24.4	0.88	1546	8.79 ± 0.12	103.58 ± 4.31
Summer	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	TEOM	21.9 ± 20.7	9.0 ± 4.7	0.74	639	2.17 ± 0.08	−2.05 ± 0.80
Winter	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	DustTrak	334.7 ± 233.4	82.7 ± 61.3	0.96	1079	2.14 ± 0.02	−8.71 ± 2.29
Summer	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	DustTrak	21.9 ± 20.7	30.5 ± 20.7	0.61	1857	0.52 ± 0.02	6.89 ± 0.59
Winter	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	GreenTek	334.7 ± 233.4	84.1 ± 54.7	0.82	798	2.66 ± 0.07	85.22 ± 6.54
Summer	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	GreenTek	21.9 ± 20.7	7.7 ± 7.6	0.79	1777	2.24 ± 0.04	6.79 ± 0.46
Winter	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	FRM filter	333.4 ± 200.1	78.0 ± 47.2	0.97	10	4.80 ± 0.45	−0.05 ± 40.30
Summer	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	FRM filter	21.9 ± 20.7	8.5 ± 3.1	0.92	13	3.48 ± 0.44	−9.87 ± 4.00
Winter	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	BAM	301.0 ± 223.3	70.0 ± 47.6	0.97	1391	4.30 ± 0.03	5.13 ± 2.48
Summer	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	BAM	22.6 ± 21.4	12.4 ± 8.2	0.86	1800	1.78 ± 0.03	−0.16 ± 0.38
Winter	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	TEOM	301.0 ± 223.3	27.7 ± 24.4	0.89	1412	8.50 ± 0.12	85.38 ± 4.13
Summer	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	TEOM	22.6 ± 21.4	9.0 ± 4.7	0.75	580	2.32 ± 0.08	−2.51 ± 0.88
Winter	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	DustTrak	301.0 ± 223.3	82.7 ± 61.3	0.96	1035	2.04 ± 0.02	−12.38 ± 2.11
Summer	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	DustTrak	22.6 ± 21.4	30.5 ± 20.7	0.64	1753	0.56 ± 0.02	6.60 ± 0.61
Winter	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	GreenTek	301.0 ± 223.3	84.1 ± 54.7	0.83	776	2.63 ± 0.06	70.07 ± 6.22
Summer	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	GreenTek	22.6 ± 21.4	7.7 ± 7.6	0.81	1718	2.34 ± 0.04	6.26 ± 0.46
Winter	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	FRM filter	316.8 ± 196.7	78.0 ± 47.2	0.97	10	4.67 ± 0.41	−5.97 ± 36.71
Summer	B_{spf}	$PM_{2.5}$	Radiance nephelometer $PM_{2.5}$	FRM filter	22.6 ± 21.4	8.5 ± 3.1	0.94	13	3.57 ± 0.40	−10.18 ± 3.59
Winter	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	$PM_{2.5}$ BAM	334.7 ± 233.4	58.4 ± 34.5	0.96	1527	4.54 ± 0.03	12.13 ± 2.69
		$PM_{10-2.5}$		$PM_{10-2.5}$ BAM[d]		9.2 ± 9.6			−0.65 ± 0.16	
Summer	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	$PM_{2.5}$ BAM	21.9 ± 20.7	9.7 ± 3.9	0.85	1891	1.63 ± 0.03	−2.26 ± 0.46
		$PM_{10-2.5}$		$PM_{10-2.5}$ BAM[d]		29.5 ± 11.1			0.13 ± 0.01	
Winter	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	$PM_{2.5}$ TEOM	334.7 ± 233.4	22.1 ± 17.7	0.84	937	8.99 ± 0.20	102.10 ± 5.50
		$PM_{10-2.5}$		$PM_{10-2.5}$ TEOM[c]		4.8 ± 6.1			−2.27 ± 0.58	
Summer	B_{sp}	$PM_{2.5}$	Radiance nephelometer TSP	$PM_{2.5}$ TEOM	21.9 ± 20.7	9.0 ± 4.8	0.74	639	2.15 ± 0.09	−2.12 ± 0.82
		$PM_{10-2.5}$		$PM_{10-2.5}$ TEOM[c]		18.2 ± 13.0			0.01 ± 0.03	
Winter	B_{sp}	B_{sp}	NGN2 nephelometer TSP	Radiance nephelometer TSP	410.4 ± 269.4	265.1 ± 181.0	0.92	1117	1.38 ± 0.02	45.62 ± 5.46
Summer	B_{sp}	B_{sp}	NGN2 nephelometer TSP	Radiance nephelometer TSP	18.8 ± 6.4	21.9 ± 20.7	0.94	2001	0.29 ± 0.00	12.52 ± 0.07
Winter	B_{sp}	B_{spf}	Radiance nephelometer TSP	Radiance nephelometer $PM_{2.5}$	333.4 ± 200.1	301.0 ± 223.3	1.00	1427	1.04 ± 0.00	5.11 ± 0.56
Summer	B_{sp}	B_{spf}	Radiance nephelometer TSP	Radiance nephelometer $PM_{2.5}$	21.9 ± 20.7	22.6 ± 21.4	1.00	2032	0.98 ± 0.00	0.16 ± 0.06

[a]Winter (December 2000 to February 2001); [b]Summer (June 2001 to August 2001); [c]B_{sp} from a Radiance Research M903 Nephelometer (Seattle, WA) without a size-selective inlet, presumably measuring total suspended particles (TSP, particles with aerodynamic diameters $< \sim 30$ μm) scattering; [d]Multiple linear regression with the B_{sp} as the dependent variable, and $PM_{2.5}$ and $PM_{10-2.5}$ concentrations as the independent variables; [e]Multiple linear regression with the B_{sp} as the dependent variable, and $PM_{2.5}$ and $PM_{10-2.5}$ concentrations as the independent variables; [f]1-hr average data were used for comparisons, except for the FRM filter mass (24-hr average). The average value was calculated over the collocated sampling period for each instrument.

exception of the desert sites, indicating that B_{sp} is a less reliable estimator of $PM_{2.5}$ during that season.

B_{sp} was compared with PM_{10} mass for a neighborhood-scale study centered on the Corcoran (COP) site from October 10, 2000, to November 14, 2000, to determine how well B_{sp} in the area is related to PM_{10}. B_{sp} and PM_{10} were not as well correlated (r = 0.66) and yielded smaller σ_{sp} (2.6 m^2/g) than those found for B_{sp} and $PM_{2.5}$ (r = 0.89, 4.5 m^2/g). This reflects the lower scattering efficiency expected for the coarse particles. IMPROVE estimates a coarse particle mass scattering efficiency (σ_{spc}) of 0.6 m^2/g.[13]

Light Scattering and PM Comparisons at FSF

B_{spf} derived from the Fresno $PM_{2.5}$ nephelometer and B_{sp} from other instruments are compared with $PM_{2.5}$ mass from different instruments in Table 3. The correlations between hourly averaged B_{spf} and $PM_{2.5}$ BAM are higher during winter (r = 0.97) than during summer (r = 0.85) with 2-fold higher $PM_{2.5}$ σ_{sp} during winter (4.3 m^2/g) than during summer (1.8 m^2/g), as shown in Figure 7. B_{sp} and B_{spf} measured with the Radiance nephelometers were highly correlated (Table 3), with a higher regression slope in winter than summer (1.04 versus 0.98). Coarse particles appear to have a small effect on the B_{sp} measurement.

Results from a multiple linear regression with B_{sp} as the dependent variable and $PM_{2.5}$ and $PM_{10-2.5}$ concentrations as the independent variables (i.e., B_{sp} = intercept + $\sigma_{spf} \times PM_{2.5}$ mass + σ_{spc} [coarse ($PM_{10-2.5}$) particle mass scattering efficiency] $\times$ $PM_{10-2.5}$ mass) are also included in Table 3. For all but the wintertime comparison with the TEOM, the σ_{spc} is negligible for these samples. $PM_{2.5}$ and

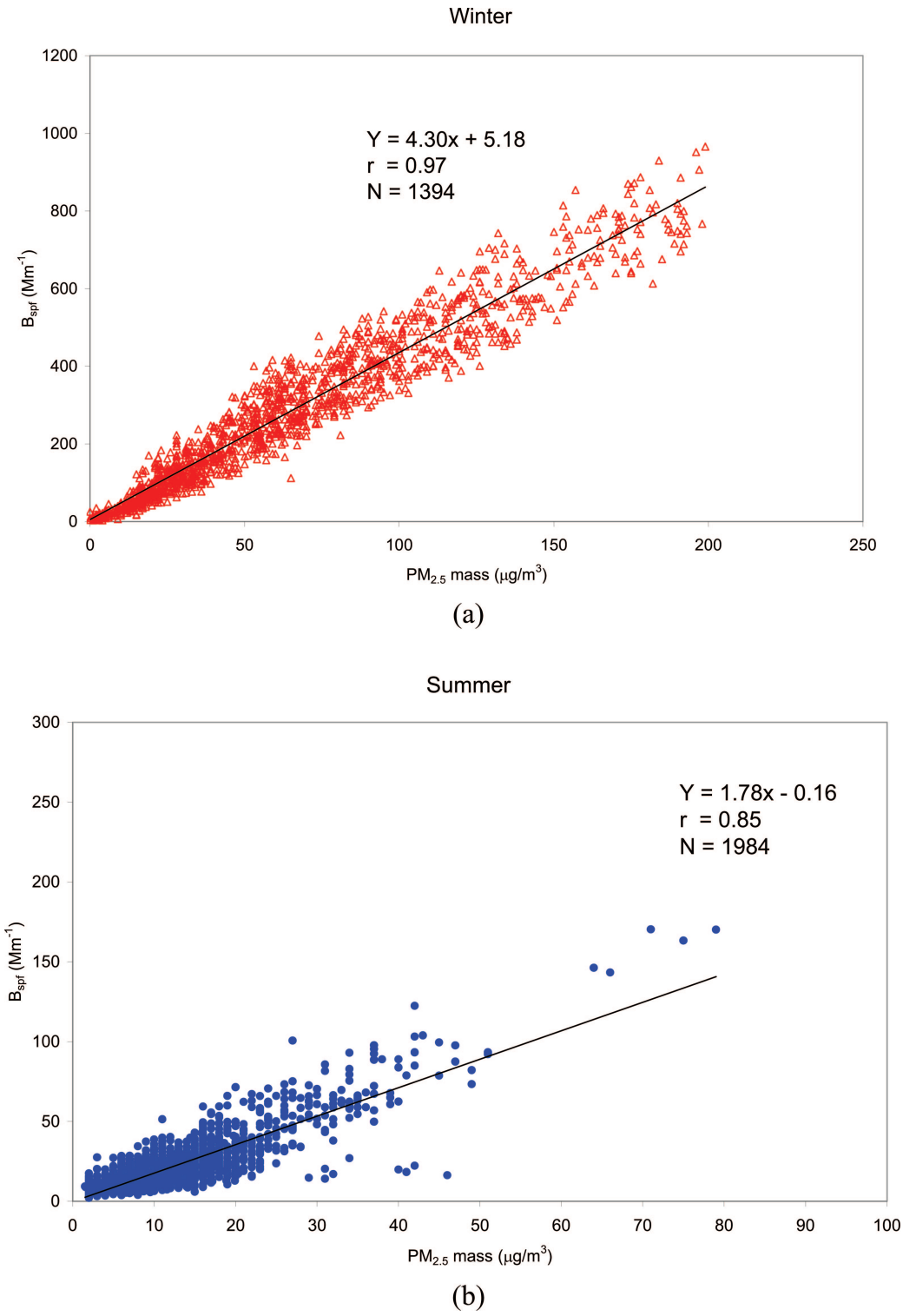


Figure 7. Comparison of hourly average B_{spf} from a Radiance Research M903 nephelometer with a $PM_{2.5}$ size-selective inlet and $PM_{2.5}$ mass determined by BAM at the FSF during (a) winter and (b) summer.

$PM_{10-2.5}$ concentrations are often correlated at Fresno, but the σ_{spf} and σ_{spc} are not necessarily constant. This results in some uncertainties, even negative values, for the multiple linear regression coefficients, especially for the TEOM measurements during summer.

On average, B_{sp} from the NGN2 ambient temperature nephelometer (410 Mm^{-1}) was 1.6 times B_{sp} from the Radiance nephelometer (265 Mm^{-1}) during the winter, which is expected, owing to the higher RH. This comparison indicates that water is being removed in the Radiance

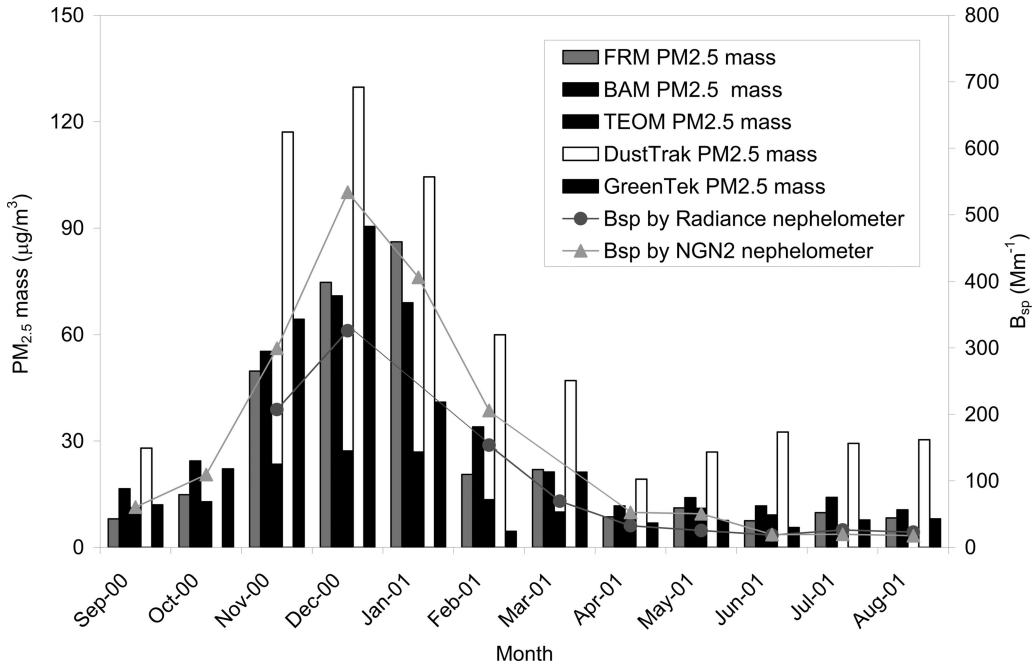


Figure 8. Monthly average B_{sp} by Radiance Research M903 and OPTEC NGN2 nephelometers and $PM_{2.5}$ mass concentrations determined by $PM_{2.5}$ FRM filter sampler, BAM, TEOM, DustTrak photometer, and GreenTek photometer at the FSF from September 2000 to August 2001.

nephelometer by the smart heater. During summer when RH was much lower than winter, average NGN2 B_{sp} (19 Mm^{-1}) was comparable to Radiance B_{sp} (22 Mm^{-1}). After excluding B_{sp} values > 1000 Mm^{-1} from the NGN2 dataset, correlations with the Radiance B_{sp} nephelometer ranged from 0.92 to 0.94.

The monthly average B_{sp} determined by the Radiance and OPTEC NGN2 nephelometers and $PM_{2.5}$ mass concentrations determined by the FRM, BAM, TEOM, DustTrak, and GreenTek are compared in Figure 8. During the winter, $PM_{2.5}$ TEOM was lower because of evaporation of NH_4NO_3 at its 50 °C internal temperature.[37-39] The filter dynamic measurement system[40] quantifies TEOM evaporation, but this was not implemented at the FSF. Among all measurements, the two photometers consistently reported higher $PM_{2.5}$ mass. On average, the DustTrak reported 2.2 and 4.3 times higher $PM_{2.5}$ mass in winter and summer, respectively, than the GreenTek.

Figures 9a and 9b show similar diurnal variations between B_{spf} and $PM_{2.5}$ mass at FSF. Although these are

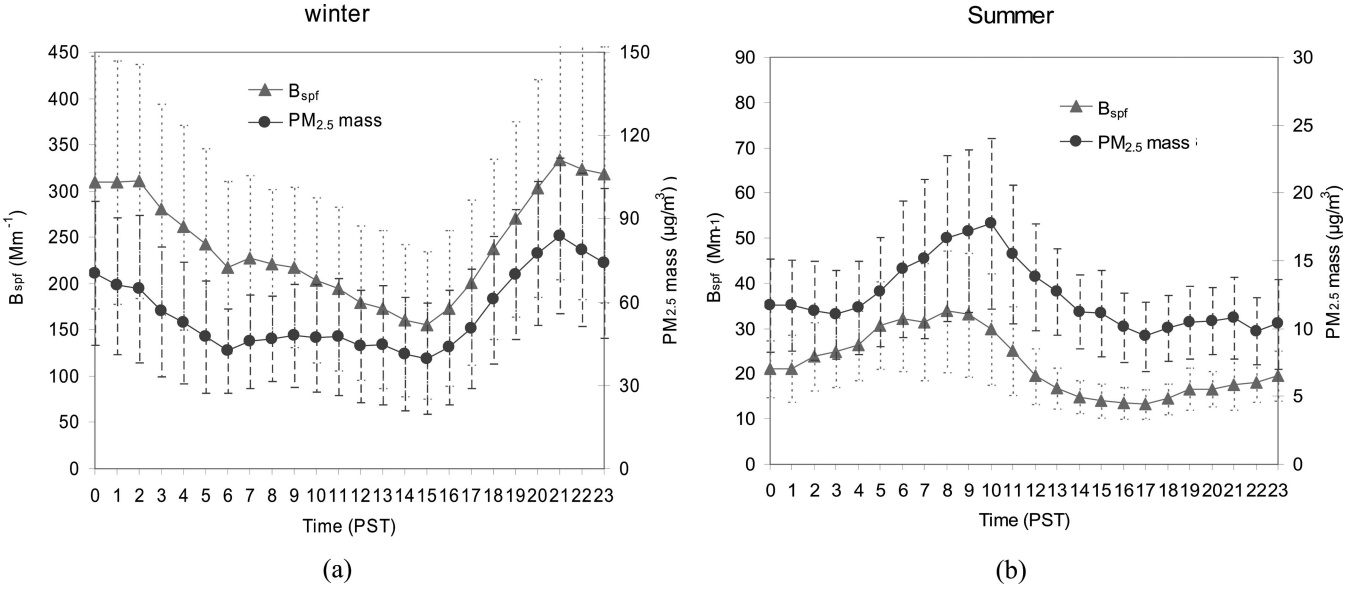


Figure 9. Diurnal variations of B_{spf} determined by the Radiance Research M903 nephelometer with a $PM_{2.5}$ size-selective inlet and $PM_{2.5}$ mass concentration determined by BAM, averaged by the time of day at the FSF during (a) winter and (b) summer.

average values, examination of individual days shows that they track each other in most situations. This comparison supports the use of $PM_{2.5}$ derived from B_{sp} to better understand $PM_{2.5}$ variability within a 24-hr period at sites that do not have all of the Supersite instrumentation.

CONCLUSION

Site-type and season-specific $PM_{2.5}$ σ_{sp} can be applied to B_{sp} measurements from a nephelometer to estimate $PM_{2.5}$ concentrations with reasonable accuracy and precision in California's San Joaquin Valley. Periodic heating of the nephelometer inlet to obtain RH below a certain value (probably <60%) can be applied to remove liquid water while minimizing evaporation of volatile compounds such as NH_4NO_3. Best agreements between B_{sp} and $PM_{2.5}$ are found during winter and for sites that are not located near sources. Summertime σ_{sp} were lower owing to drier conditions with less hygroscopic NH_4NO_3 and a larger proportion of soil-related PM, all of which result in lower σ_{sp}.

ACKNOWLEDGMENTS

This work was supported by the CRPAQS under the management of the California Air Resources Board and U.S. Environmental Protection Agency under Contract #R-82805701 for the Fresno Supersite. The conclusions are those of the authors and do not necessarily reflect the views of the sponsoring agencies. Any mention of commercially available products and supplies does not constitute an endorsement of these products and supplies. The authors thank Jo Gerrard and Tim Richard of DRI for assembling and editing the manuscript.

REFERENCES

1. Beuttell, R.G.; Brewer, A.W. Instruments for the Measurement of the Visual Range; *J. Sci. Instrum.* **1949**, *26,* 357-359.
2. Ahlquist, N.C.; Charlson, R.J. A New Instrument for Evaluating the Visual Quality of Air; *J. Air Pollut. Control Assoc.* **1967**, *17,* 467-469.
3. Ruby, M.G.; Rood, M.J.; Chow, J.C.; Egami, R.T.; Rogers, C.F.; Watson, J.G. Integrating Nephelometer Measurement of Scattering Coefficient and Fine Particle Concentrations. In *Methods of Air Sampling and Analysis*, 3rd ed.; Lodge, J.P., Ed.; Lewis: Chelsea, MI, 1989; pp. 450-457.
4. Brauer, M. Assessment of Indoor Aerosols with an Integrating Nephelometer; *J. Expo. Anal. Environ. Epidemiol.* **1995**, *5,* 45-56.
5. Charlson, R.J.; Ahlquist, N.C.; Horvath, H. On the Generality of Correlation of Atmospheric Aerosol Mass Concentration and Light Scatter; *Atmos. Environ.* **1968**, *2,* 455-464.
6. Chow, J.C.; Watson, J.G.; Lowenthal, D.H.; Richards, L.W. Comparability between $PM_{2.5}$ and Light Scattering Measurements; *Environmental Monitoring and Assessment,* **2002**, *79,* 29-45.
7. Lewis, C.W. On the Proportionality of Fine Mass Concentration and Extinction Coefficient for Bimodal Size Distributions; *Atmos. Environ.* **1981**, *15,* 2639-2646.
8. Scheff, P.A.; Wadden, R.A. Comparisons of Three Methods of Particulate Measurements in Chicago Air; *Atmos. Environ.* **1979**, *16*, 639-643.
9. Sioutas, C.; Kim, S.; Chang, M.C.; Terrell, L.L.; Gong, H. Field evaluation of a modified DataRAM MIE Scattering Monitor for Real-Time $PM_{2.5}$ Mass Concentration Measurements; *Atmos. Environ.* **2000**, *34,* 4829-4838.
10. Thielke, J.F.; Charlson, R.J.; Winter, J.W.; Ahlquist, N.C.; Whitby, K.T.; Husar, R.B.; Liu, B.Y.H. Multiwavelength Nephelometer Measurements in Los Angeles Smog Aerosols. II. Correlation with Size Distributions, Volume Concentrations; *J. Colloid Interface Sci.* **1972**, *39,* 252-259.
11. Thomas, A.; Gebhart, J. Correlations between Gravimetry and Light Scattering Photometry for Atmospheric Aerosols; *Atmos. Environ.* **1994**, *28,* 935-938.
12. Waggoner, A.P.; Weiss, R.E. Comparison of Fine Particle Mass Concentration and Light Scattering Extinction in Ambient Aerosol; *Atmos. Environ.* **1980**, *14,* 623-626.
13. Watson, J.G. Visibility: Science and Regulation; *J. Air & Waste Manage. Assoc.* **2002**, *52,* 628-713.
14. Ensor, D.S.; Waggoner, A.P. Angular Truncation Error in the Integrating Nephelometer; *Atmos. Environ.* **1970**, *4,* 481-487.
15. Ensor, D.S.; Pilat, M.J. The Effect of Particle Size Distribution on Light Transmittance Measurement; *J. Am. Ind. Hyg. Assoc.* **1971**, *32,* 287-292.
16. Guyon, P.; Graham, B.; Beck, J.; Boucher, O.; Gerasopoulos, E.; Mayol-Bracero, O.L.; Roberts, G.C.; Artaxo, P.; Andreae, M.O. Physical Properties and Concentrations of Aerosol Particles Over the Amazon Tropical Forest During Background and Biomass Burning Conditions; *Atmos. Chem. Physics Discussion,* **2003**, *3,* 1333-1366.
17. Moosmüller, H.; Arnott, W.P. Angular Truncation Errors in Integrating Nephelometry; *Rev. Sci. Instrum.* **2003**, *74,* 3492-3501, doi:10.1063/1.1581355.
18. Quenzel, H.; Ruppersberg, G.H.; Schellhase, R. Calculations about the Systematic Error of Visibility-Meters Measuring Scattered Light; *Atmos. Environ.* **1975**, *9,* 587-601.
19. Rabinoff, R.A.; Herman, B.M. Effect of Aerosol Size Distribution on the Accuracy of the Integrating BNephelometer; *J. Appl. Meteorol.* **1973**, *12,* 184-186.
20. Rosen, J.M.; Pinnick, R.G.; Garvey, D.M. Nephelometer Optical Response Model for the Interpretation of Atmospheric Aerosol Measurements; *Appl. Opt.* **1997**, *36,* 2642-2649.
21. Varma, R.; Moosmüller, H.; Arnott, W.P. Toward an Ideal Integrating Nephelometer; *Opt. Lett.* **2003**, *28,* 1007-1009.
22. Watson, J.G.; DuBois, D.W.; DeMandel, R.; Kaduwela, A.P.; Magliano, K.L.; McDade, C.; Mueller, P.K.; Ranzieri, A.J.; Roth, P.M.; Tanrikulu, S. *Field Program Plan for the California Regional $PM_{2.5}$/PM_{10} Air Quality Study (CRPAQS*); Prepared for California Air Resources Board, Sacramento, CA, by Desert Research Institute: Reno, NV, 1998.
23. Richards, L.W.; Weiss, R.E.; Waggoner, A.P. Radiance Research Model 903 Integrating Nephelometer. In *Proceedings, Regional Haze and Global Radiation Balance–Aerosol Measurements and Models: Closure, Reconciliation and Evaluation*; A&WMA: Pittsburgh, PA, 2001, 3B7-1–3B7-5.
24. Richards, L.W.; Lehrman, D.E.; Weiss, R.E.; Bush, D.; Watson, J.G.; McDade, C.E.; Magliano, K. Light Scattering Measurements during the California Regional PM_{10}/$PM_{2.5}$ Air Quality Study. In *Proceedings, Regional Haze and Global Radiation Balance - Aerosol Measurements and Models: Closure, Reconciliation and Evaluation*; A&WMA: Pittsburgh, PA, 2001, 5A2-1–5A2-5.
25. Alcorn, S.H.; Richards, L.W.; Lehrman, D.E. *Comparisons Between Light Scattering and Fine Particle Mass Data;* STI-90232102644; Prepared for California Air Resources Board, Sacramento, CA, by Sonoma Technology, Inc.: Petaluma, CA, 2004.
26. Baldauf, R.W.; Lane, D.D.; Marotz, G.A.; Wiener, R.W. Performance evaluation of the Portable MiniVol Particulate Matter Sampler; *Atmos. Environ.* **2001**, *35,* 6087-6091.
27. Watson, J.G.; Chow, J.C.; Bowen, J.L.; Lowenthal, D.H.; Hering, S.; Ouchida, P.; Oslund, W. Air quality Measurements from the Fresno Supersite; *J. Air & Waste Manage. Assoc.* **2000**, *50,* 1321-1334.
28. Chow, J.C.; Watson, J.G.; Lowenthal, D.H.; Solomon, P.A.; Magliano, K.L.; Ziman, S.D.; Richards, L.W. PM_{10} and $PM_{2.5}$ Compositions in California's San Joaquin Valley; *Aerosol Sci. Technol.* **1993**, *18,* 105-128.
29. Watson, J.G.; Chow, J.C.; Park, K.; Lowenthal, D.H. Nanoparticle and Ultrafine Particle Events at the Fresno Supersite; *J. Air & Waste Manage. Assoc.* in press.
30. Chow, J.C.; Watson, J.G.; Lowenthal, D.H.; Solomon, P.A.; Magliano, K.L.; Ziman, S.D.; Richards, L.W. PM_{10} Source Apportionment in California's San Joaquin Valley; *Atmos. Environ.* **1992**, *26A,* 3335-3354.
31. Chow, J.C.; Watson, J.G.; Lu, Z.; Lowenthal, D.H.; Frazier, C.A.; Solomon, P.A.; Thuillier, R.H.; Magliano, K.L. Descriptive analysis of $PM_{2.5}$ and PM_{10} at Regionally Representative Locations during SJVAQS/AUSPEX; *Atmos. Environ.* **1996**, *30,* 2079-2112.
32. Chow, J.C.; Watson, J.G.; Lowenthal, D.H.; Hackney, R.; Magliano, K.L.; Lehrman, D.; Smith, T. Temporal Variations of $PM_{2.5}$, PM_{10}, and Gaseous Precursors during the 1995 Integrated Monitoring Study in Central California; *J. Air & Waste Manage. Assoc.***1999**, *49(PM),* PM16-PM24.
33. Watson, J.G.; Chow, J.C. A Wintertime $PM_{2.5}$ Episode at the Fresno, CA Supersite; *Atmos. Environ.* **2002**, *36,* 465-475.
34. Chow, J.C.; Watson, J.G.; Chen, L.W.A. $PM_{2.5}$ Chemical Composition and Spatiotemporal Variability during the California Regional PM_{10}/$PM_{2.5}$ Air Quality Study (CRPAQS); *J. Geophys. Res.* in press.
35. White, W.H.; Macias, E.S. Chemical Mass Balancing with Ill Defined Sources: Regional Apportionment in the California Desert; *Atmos. Environ.* **1991**, *25A,* 1547-1557.
36. Green, M.C.; Flocchini, R.G.; Myrup, L.O. The relationship of the Extinction Coefficient Distribution to Wind FIeld Patterns in Southern California; *Atmos. Environ.* **1992**, *26A,* 827-840.

37. Jaques, P.A.; Ambs, J.L.; Grant, W.L.; Sioutas, C. Field Evaluation of the Differential TEOM Monitor for Continuous $PM_{2.5}$ Mass Concentrations; *Aerosol Sci. Technol.* **2004**, *38(Suppl. 1),* 49-59.
38. Hitzenberger, R.; Berner, A.; Galambos, Z.; Maenhaut, W.; Cafmeyer, J.; Schwarz, J.; Muller, K.; Spindler, G.; Wieprecht, W.; Acker, K.; Hillamo, R.; Makela, T. Intercomparison of Methods to Measure the Mass Concentration of the Atmospheric Aerosol during INTERCOMP2000–Influence of Instrumentation and Size Cuts; *Atmos. Environ.* **2004**, *38,* 6467-6476.
39. Hering, S.; Fine, P.M.; Sioutas, C.; Jaques, P.A.; Ambs, J.L.; Hogrefe, O.; Demerjian, K.L. Field assessment of the Dynamics of Particulate Nitrate Vaporization Using Differential TEOM (R) and Automated Nitrate Monitors; *Atmos. Environ.* **2004**, *38,* 5183-5192.
40. Grover, B.D.; Kleinman, M.; Eatough, N.L.; Eatough, D.J.; Hopke, P.K.; Long, R.W.; Wilson, W.E.; Meyer, M.B.; Ambs, J.L. Measurement of total $PM_{2.5}$ Mass (Nonvolatile Plus Semivolatile) with the Filter Dynamic Measurement System Tapered Element Oscillating Microbalance Monitor; *J. Geophys. Res.* **2005**, *110(D7),* D07S03, doi:10.1029/2004JD004995.

About the Authors

Judith C. Chow and John G. Watson are Research Professors, and Douglas H. Lowenthal and Norman F. Robinson are Associate Research Professors, with the Desert Research Institute (DRI). Kihong Park was formerly an Assistant Research Professor at DRI and is currently an Assistant Professor at Gwangju Institute of Science and Technology. Karen A. Magliano is Chief of the Air Quality Data Branch at the California Air Resouces Board. Address correspondence to: Judith C. Chow, Division of Atmospheric Sciences, Desert Research Institute, 2215 Raggio Parkway, Reno, NV 89512 ; phone: +1-775-674-7050; fax: 1-775-674-7009; e-mail: judyc@dri.edu.